Synthetic and YouTube Combination Results

Threshold 0.3

1. Review Count Effect: No Balancing Method

Corpus	Rows	Perfumes	LR F1	XGBoost F1
Synthetic 784x3	99	33	Mean 0.5931 SD 0.0491	Mean 0.5830 SD 0.0906
	201	67	Mean 0.5998 SD 0.0276	Mean 0.5877 SD 0.0533
	300	100	Mean 0.6043 SD 0.0141	Mean 0.6087 SD 0.0202
	501	167	Mean 0.6320 SD 0.0389	Mean 0.6055 SD 0.0140
	1002	334	Mean 0.6385 SD 0.0142	Mean 0.6251 SD 0.0157
	2001	667	Mean 0.6307 SD 0.0072	Mean 0.6318 SD 0.0095
	2352	784	**Mean 0.6514** SD 0.0000	**Mean 0.6576** SD 0.0056
Synthetic 784x6	198	33	Mean 0.5524 SD 0.0622	Mean 0.5495 SD 0.0675
	402	67	Mean 0.5556 SD 0.0264	Mean 0.5494 SD 0.0173
	600	100	Mean 0.5744 SD 0.0167	Mean 0.5638 SD 0.0282
	1002	167	Mean 0.5790 SD 0.0112	Mean 0.5616 SD 0.0096
	2004	334	Mean 0.5874 SD 0.0079	Mean 0.5596 SD 0.0124
	4002	667	Mean 0.5829 SD 0.0054	Mean 0.5797 SD 0.0052
	4704	784	**Mean 0.5876** SD 0.0000	**Mean 0.5800** SD 0.0018
YouTube only	784	784	**0.3964**	**0.3601**
YouTube + synthetic 784x3	3136	784	**0.4131**	**0.3609**
YouTube + synthetic 784x6	5488	784	**0.3526**	**0.4171**

No balancing method; mixed rows are evaluated on YouTube rows.

Emotion-count balancing

Main question: balance emotion counts and compare F1.

2.0 Summary: No Balancing vs Hybrid vs SMOTE

Training data	Logistic Regression			XGBoost		
	No balancing	Hybrid	SMOTE	No balancing	Hybrid	SMOTE
Synthetic 784x3	**0.6514**	0.6022	0.6358	0.6576	0.6130	**0.6577**
Synthetic 784x6	**0.5876**	0.5456	0.5731	0.5800	0.5472	**0.5967**
YouTube only	**0.3964**	0.2735	0.3812	0.3601	0.2346	**0.3695**
YouTube + synthetic 784x3	**0.4131**	0.3323	0.3787	0.3609	**0.3684**	0.3445
YouTube + synthetic 784x6	0.3526	0.3483	**0.4051**	**0.4171**	0.4058	0.3662

Best F1 in each row/model is shaded.

2.1 No balancing method: Reference

Training data	LR F1	XGBoost F1
YouTube only	**0.3964**	0.3601
YouTube + synthetic 784x3	**0.4131**	0.3609
YouTube + synthetic 784x6	0.3526	**0.4171**

Emotion counts unchanged.

2.2 oversampling: Cross-emotion count balancing

Training data	LR F1	LR change	XGBoost F1	XGBoost change
YouTube only	0.3760	-0.0205	**0.3709**	+0.0108
YouTube + synthetic 784x3	0.3644	-0.0487	**0.3685**	+0.0075
YouTube + synthetic 784x6	0.3566	+0.0040	0.3554	-0.0617

Raises low emotion categories.

2.3 undersampling: Cross-emotion count balancing

Training data	LR F1	LR change	XGBoost F1	XGBoost change
YouTube only	0.2660	-0.1304	0.2503	-0.1098
YouTube + synthetic 784x3	0.3835	-0.0296	0.3467	-0.0142
YouTube + synthetic 784x6	**0.3901**	+0.0375	0.4061	-0.0110

Lowers high emotion categories.

2.4 hybrid (undersampling + oversampling): Cross-emotion count balancing

Training data	LR F1	LR change	XGBoost F1	XGBoost change
YouTube only	0.2735	-0.1229	0.2346	-0.1256
YouTube + synthetic 784x3	0.3323	-0.0808	0.3684	+0.0075
YouTube + synthetic 784x6	0.3483	-0.0043	0.4058	-0.0113

Lowers high, raises low categories.

2.5 SMOTE: Binary-label reference

Training data	LR F1	LR change	XGBoost F1	XGBoost change
YouTube only	0.3812	-0.0152	0.3695	+0.0094
YouTube + synthetic 784x3	0.3787	-0.0344	0.3445	-0.0164
YouTube + synthetic 784x6	0.4051	+0.0524	0.3662	-0.0508

Binary SMOTE reference.

Balance achieved

Show the actual emotion counts after each balancing method.

3. Mean Emotion Counts After Balancing

Training data	Method	Rows	Count SD	Joy	Sadness	Fear	Trust	Surprise	Anticipation	Disgust
YouTube only	No balancing method	627	193.8	498	9	2	381	35	293	15
	Oversampling	706	187.1	498	35	35	381	36	326	35
	Undersampling	108	14.0	39	9	2	35	35	35	15
	hybrid (undersampling + oversampling)	175	**8.5**	52	35	35	36	48	56	35
	SMOTE	label-wise	112.5	498	618	625	381	592	334	612
YouTube + synthetic 784x3	No balancing method	2509	568.9	747	136	179	1048	654	1760	34
	Oversampling	3857	376.0	1013	654	654	1048	922	1795	654
	Undersampling	1625	305.2	654	136	179	654	654	876	34
	hybrid (undersampling + oversampling)	2978	**131.3**	916	654	654	655	918	926	654
	SMOTE	label-wise	358.5	1762	2373	2330	1461	1854	1760	2475
YouTube + synthetic 784x6	No balancing method	4390	825.9	845	175	430	1700	1183	2558	110
	Oversampling	6034	591.1	1022	845	845	1700	1360	2558	845
	Undersampling	2659	316.0	845	175	430	845	845	865	110
	hybrid (undersampling + oversampling)	4257	**98.5**	1067	845	845	846	1066	865	845
	SMOTE	label-wise	650.6	3546	4215	3961	2690	3207	2558	4281

Lower Count SD means better balance within the same training setup.

Binary emotion balancing

Balance each emotion as positive vs negative; SMOTE belongs in this group.

4.1 No balancing method: Binary-label results by emotion

Training data	Emotion	Train emotion count	LR F1	XGBoost F1
YouTube only	joy	498	0.8815	0.8902
	trust	381	0.7426	0.7368
	anticipation	293	0.6275	0.5947
	surprise	35	0.1600	0.0385
	sadness	9	0.1053	0.0000
	disgust	15	0.2581	0.2609
	fear	2	0.0000	0.0000
YouTube + synthetic 784x3	joy	747	0.8734	0.8913
	trust	1048	0.7429	0.7438
	anticipation	1760	0.6249	0.6096
	surprise	654	0.2222	0.1000
	sadness	136	0.0000	0.0000
	disgust	34	0.1429	0.1818
	fear	179	0.2857	0.0000
YouTube + synthetic 784x6	joy	845	0.8730	0.8889
	trust	1700	0.7344	0.7472
	anticipation	2558	0.6290	0.6155
	surprise	1183	0.1579	0.0678
	sadness	175	0.0000	0.0000
	disgust	110	0.0741	0.1000
	fear	430	0.0000	0.5000

No rows changed.

4.2 oversampling: Binary-label results by emotion

Training data	Emotion	Train emotion count	LR F1	LR change	XGBoost F1	XGBoost change
YouTube only	joy	498	0.8591	-0.0224	0.8839	-0.0063
	trust	381	0.7299	-0.0127	0.7214	-0.0154
	anticipation	334	0.6348	+0.0073	0.5987	+0.0040
	surprise	592	0.1605	+0.0005	0.0882	+0.0498
	sadness	618	0.0392	-0.0660	0.0000	+0.0000
	disgust	612	0.1136	-0.1444	0.2400	-0.0209
	fear	625	0.0909	+0.0909	0.0000	+0.0000
YouTube + synthetic 784x3	joy	1762	0.8841	+0.0106	0.8894	-0.0019
	trust	1461	0.7453	+0.0024	0.7494	+0.0055
	anticipation	1760	0.5063	-0.1185	0.5690	-0.0405
	surprise	1854	0.2609	+0.0386	0.0923	-0.0077
	sadness	2373	0.0136	+0.0136	0.0000	+0.0000
	disgust	2475	0.0949	-0.0480	0.2857	+0.1039
	fear	2330	0.1429	-0.1429	0.0000	+0.0000
YouTube + synthetic 784x6	joy	3546	0.8857	+0.0127	0.8882	-0.0006
	trust	2690	0.7483	+0.0139	0.7492	+0.0020
	anticipation	2558	0.6187	-0.0103	0.6117	-0.0039
	surprise	3207	0.2069	+0.0490	0.1053	+0.0375
	sadness	4215	0.0229	+0.0229	0.0000	+0.0000
	disgust	4281	0.0692	-0.0049	0.2143	+0.1143
	fear	3961	0.2222	+0.2222	0.0000	-0.5000

Duplicates minority class per emotion.

4.3 undersampling: Binary-label results by emotion

Training data	Emotion	Train emotion count	LR F1	LR change	XGBoost F1	XGBoost change
YouTube only	joy	130	0.8293	-0.0522	0.7987	-0.0915
	trust	246	0.7254	-0.0172	0.6729	-0.0638
	anticipation	293	0.6277	+0.0002	0.6035	+0.0088
	surprise	35	0.1232	-0.0368	0.1385	+0.1000
	sadness	9	0.0189	-0.0864	0.0200	+0.0200
	disgust	15	0.0543	-0.2038	0.0481	-0.2127
	fear	2	0.0084	+0.0084	0.0076	+0.0076
YouTube + synthetic 784x3	joy	747	0.8849	+0.0114	0.8875	-0.0038
	trust	1048	0.7494	+0.0065	0.7534	+0.0096
	anticipation	749	0.4986	-0.1263	0.4555	-0.1540
	surprise	654	0.2452	+0.0229	0.1379	+0.0379
	sadness	136	0.0214	+0.0214	0.0000	+0.0000
	disgust	34	0.0580	-0.0849	0.0718	-0.1100
	fear	179	0.0833	-0.2024	0.5000	+0.5000
YouTube + synthetic 784x6	joy	845	0.8798	+0.0068	0.8878	-0.0011
	trust	1700	0.7442	+0.0098	0.7570	+0.0099
	anticipation	1832	0.6172	-0.0118	0.5819	-0.0337
	surprise	1183	0.2222	+0.0643	0.1538	+0.0860
	sadness	175	0.0218	+0.0218	0.0315	+0.0315
	disgust	110	0.0458	-0.0282	0.0627	-0.0373
	fear	430	0.0571	+0.0571	0.2500	-0.2500

Drops majority class per emotion.

4.4 SMOTE: Binary-label results by emotion

Training data	Emotion	Train emotion count	LR F1	LR change	XGBoost F1	XGBoost change
YouTube only	joy	498	0.8493	-0.0322	0.8714	-0.0188
	trust	381	0.7284	-0.0143	0.7180	-0.0187
	anticipation	334	0.6329	+0.0054	0.5898	-0.0049
	surprise	592	0.1754	+0.0154	0.1373	+0.0988
	sadness	618	0.0388	-0.0664	0.0000	+0.0000
	disgust	612	0.1258	-0.1323	0.2703	+0.0094
	fear	625	0.1176	+0.1176	0.0000	+0.0000
YouTube + synthetic 784x3	joy	1762	0.8817	+0.0082	0.8915	+0.0002
	trust	1461	0.7508	+0.0080	0.7438	+0.0000
	anticipation	1760	0.5103	-0.1145	0.5430	-0.0666
	surprise	1854	0.2361	+0.0139	0.0513	-0.0487
	sadness	2373	0.0000	+0.0000	0.0000	+0.0000
	disgust	2475	0.0901	-0.0528	0.1818	+0.0000
	fear	2330	0.1818	-0.1039	0.0000	+0.0000
YouTube + synthetic 784x6	joy	3546	0.8839	+0.0110	0.8881	-0.0008
	trust	2690	0.7450	+0.0106	0.7510	+0.0039
	anticipation	2558	0.6131	-0.0159	0.5975	-0.0180
	surprise	3207	0.2174	+0.0595	0.1778	+0.1100
	sadness	4215	0.0290	+0.0290	0.0000	+0.0000
	disgust	4281	0.0803	+0.0062	0.1493	+0.0493
	fear	3961	0.2667	+0.2667	0.0000	-0.5000

Creates minority samples per emotion.

Synthetic-only balancing

Apply binary balancing methods when training and testing only on synthetic reviews.

5.1 No balancing method: Synthetic-only

Corpus	LR F1	XGBoost F1
Synthetic 784x3	**Mean 0.6514** SD 0.0000	Mean 0.6576 SD 0.0056
Synthetic 784x6	**Mean 0.5876** SD 0.0000	Mean 0.5800 SD 0.0018

No rows changed.

5.2 oversampling: Synthetic-only

Corpus	LR F1	LR change	XGBoost F1	XGBoost change
Synthetic 784x3	Mean 0.6319 SD 0.0013	-0.0195	Mean 0.6581 SD 0.0064	+0.0004
Synthetic 784x6	Mean 0.5645 SD 0.0008	-0.0231	Mean 0.5954 SD 0.0026	+0.0154

Duplicates minority class per emotion.

5.3 undersampling: Synthetic-only

Corpus	LR F1	LR change	XGBoost F1	XGBoost change
Synthetic 784x3	Mean 0.5946 SD 0.0055	-0.0567	Mean 0.6168 SD 0.0033	-0.0408
Synthetic 784x6	Mean 0.5507 SD 0.0014	-0.0369	Mean 0.5706 SD 0.0016	-0.0094

Drops majority class per emotion.

5.4 SMOTE: Synthetic-only

Corpus	LR F1	LR change	XGBoost F1	XGBoost change
Synthetic 784x3	Mean 0.6358 SD 0.0013	-0.0155	Mean 0.6577 SD 0.0020	+0.0001
Synthetic 784x6	Mean 0.5731 SD 0.0007	-0.0145	Mean 0.5967 SD 0.0019	+0.0167

Creates minority samples per emotion.

Best LR count balance for YouTube + synthetic 784x6: hybrid (undersampling + oversampling), Count SD 98.5, LR F1 0.3483.

Best LR F1 among balancing methods: YouTube + synthetic 784x6 + undersampling, Macro F1 0.3901.

Best LR overall shown: YouTube + synthetic 784x3 + no balancing method, Macro F1 0.4131.